

Centered Giambelli Compatibility Does Not Characterize Determinantal Point Processes

Automatic mathematics run `fa_banach_001`

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Status. Candidate counterexample, likely valid, for expert review. For every $R > 0$, the simple point process consisting of two independent $\text{Cauchy}(0, R)$ points satisfies every hypothesis of Conjecture 1 in Bufetov–Lazag but is not determinantal. Thus the conjecture as stated has a full negative answer.

1 Source conjecture

The source is A. I. Bufetov and P. Lazag, *A determinantal point process governed by an integrable projection kernel is Giambelli compatible*, arXiv:2111.05606 [1]. For a point configuration $X \subset \mathbb{R}$ and $R > 0$, its centered specialization is prescribed on the Newton power sums by

$$\rho^R(p_k)(X) = \sum_{x \in X} \frac{1}{(x + iR)^k} - \mathbb{E}_{\mathbb{P}} \sum_{x \in X} \frac{1}{(x + iR)^k}, \quad k \geq 1. \quad (1)$$

On printed page 5, Conjecture 1 asks whether every point process for which ρ^R is an L^1 -specialization and $(\mathbb{P}, \rho^R)$ is Giambelli compatible must be determinantal. The source statement is reproduced below.

Examples of determinantal point processes with an integrable projection kernel include the sine process, the Airy process, the Bessel process (see [14]), as well as determinantal point processes governed by reproducing kernels of de Branges spaces (see [20]). In the discrete setting, the discrete sine process ([9]), the process with the Gamma kernel ([10]) and the process with the discrete Bessel kernel ([6]) also form examples.

See for example [7] for a survey on determinantal point processes.

Borodin, Olshanski and Strahov [11] (see also Borodin–Strahov [13]) derive the determinantal property from Giambelli compatibility.

Conjecture 1. *Let $\mathbb{P}$ be a point process on E such that ρ^R is an L^1 -specialization. The pair $(\mathbb{P}, \rho^R)$ is Giambelli compatible only if $\mathbb{P}$ is a determinantal point process.*

For further background on the use of symmetric functions in the theory of point processes, see e.g. [34], [36], [35], [27], [12], [8], [21], while for further background on characteristic polynomials of random matrices, one can see [28], [24], [41], [3], or, more recently, [22], [13], [11], [5], [30] for the discrete case.

Here Giambelli compatibility means that, for every partition $\lambda = (p_1, \dots, p_d \mid q_1, \dots, q_d)$ in Frobenius notation,

$$\mathbb{E}_{\mathbb{P}}[\rho^R(s_{\lambda})] = \det(\mathbb{E}_{\mathbb{P}}[\rho^R(s_{(p_i|q_j)})])_{i,j=1}^d. \quad (2)$$

2 Proof intuition

The centering in (1) creates a degeneracy on harmonic measure. The Möbius map $x \mapsto (x + iR)^{-1}$ sends Cauchy(0, R) measure to uniform arclength on a circle. Holomorphic mean value then replaces each random transformed point by the center of that circle. With two independent points, centering subtracts exactly the power sums of those two center points. Consequently, the expectation of the entire centered symmetric-function specialization is evaluation at the zero alphabet. Every nonempty Schur expectation vanishes, so all Giambelli identities hold for a reason unrelated to fermionic or determinantal structure.

To rule out a hidden determinantal representation, use only the first four correlation functions. A fixed-two iid process has correlation densities 2, 2, 0, 0. Were these principal minors of a kernel, the two- and three-minor equations would force a normalized four-point kernel matrix into one of two tournament forms. The only form allowed by the three-minors has determinant -4 , whereas the four-point correlation requires determinant zero.

3 Counterexample

Theorem 1 (Iid Cauchy counterexample). *Fix $R > 0$, and let ν_R be the Cauchy probability measure*

$$d\nu_R(x) = \frac{R}{\pi(x^2 + R^2)} dx \quad (x \in \mathbb{R}). \quad (3)$$

Let Y_1, Y_2 be independent with law ν_R , and let $\mathbb{P}$ be the law on $\text{Conf}(\mathbb{R})$ of the random set $X = \{Y_1, Y_2\}$. Then:

1. *X is a simple two-point configuration almost surely;*
2. *the specialization ρ^R in (1) is an $L^1(\text{Conf}(\mathbb{R}), \mathbb{P})$ -specialization;*
3. *$(\mathbb{P}, \rho^R)$ is Giambelli compatible;*
4. *$\mathbb{P}$ is not a determinantal point process, even if complex-valued, nonsymmetric measurable kernels and arbitrary dominating reference measures are allowed.*

Hence Conjecture 1 of [1] is false.

4 Proof

Because ν_R is atomless, $Y_1 \neq Y_2$ almost surely. Thus the unordered set X is indeed a simple point configuration with exactly two points.

Cauchy measure becomes uniform circle measure

Put

$$T(x) = \frac{1}{x + iR}, \quad c = -\frac{i}{2R}, \quad r = \frac{1}{2R}. \quad (4)$$

If Θ is uniform on $(0, 2\pi)$, then $R \cot(\Theta/2)$ has law ν_R , and a direct calculation gives

$$T(R \cot(\Theta/2)) = -\frac{i}{2R} + \frac{ie^{-i\Theta}}{2R}. \quad (5)$$

Thus $T_{\#}\nu_R$ is uniform arclength on the circle $\{c + r\zeta : |\zeta| = 1\}$. In particular, the one-variable mean-value identity gives

$$\int_{\mathbb{R}} T(x)^k d\nu_R(x) = c^k, \quad k \geq 1. \quad (6)$$

Let $t_j = T(Y_j)$. More generally, for every polynomial $F \in \mathbb{C}[z_1, z_2]$, independence and two applications of the mean-value identity give

$$\mathbb{E}F(t_1, t_2) = F(c, c). \quad (7)$$

L^1 specialization and Giambelli compatibility

Since $|t_j| \leq R^{-1}$, (6) yields

$$\rho^R(p_k) = t_1^k + t_2^k - 2c^k, \quad k \geq 1. \quad (8)$$

The algebra Λ of symmetric functions is the polynomial algebra $\mathbb{C}[p_1, p_2, \dots]$. The image under ρ^R of any element of Λ is therefore a bounded random variable. This proves the required L^1 -specialization property.

Let $f \in \Lambda$. After substituting (8), the random variable $\rho^R(f)$ is an ordinary polynomial in t_1, t_2 . Applying (7) evaluates that polynomial at $t_1 = t_2 = c$. At this point every power sum in (8) becomes zero, and hence

$$\mathbb{E}[\rho^R(f)] = f|_{p_1=p_2=\dots=0}. \quad (9)$$

For every nonempty partition λ , the Schur function s_λ is homogeneous of positive degree, so

$$\mathbb{E}[\rho^R(s_\lambda)] = 0. \quad (10)$$

All hook partitions in (2) are also nonempty. Both sides of (2) are consequently zero for every nonempty λ ; the empty partition gives $1 = 1$. This proves Giambelli compatibility.

The iid two-point process is not determinantal

Relative to the reference probability measure ν_R , the ordered factorial correlation densities of X are

$$\rho_1(x) = 2, \quad \rho_2(x, y) = 2 \quad (x \neq y), \quad \rho_m = 0 \quad (m \geq 3). \quad (11)$$

Indeed, the two labels give two choices for one ordered point and two choices for an ordered pair, while no ordered tuple of three distinct points exists.

It suffices to rule out a kernel relative to ν_R . To see that this also rules out every other reference measure, suppose that a kernel K represents the correlation measures relative to a measure μ . Since $\rho_1 = 2\nu_R$, we have $\nu_R \ll \mu$; write $h = d\nu_R/d\mu$. On the ν_R -conull set where $h > 0$, the rescaled kernel

$$B(x, y) = \frac{K(x, y)}{\sqrt{h(x)h(y)}} \quad (12)$$

represents the same correlations relative to ν_R , because every $m \times m$ determinant is divided by $\prod_{j=1}^m h(x_j)$.

Assume for contradiction that such a measurable B exists. By (11), for almost every tuple of pairwise distinct points, its principal minors satisfy

$$\det B_{\{j\}} = 2, \quad \det B_{\{j, k\}} = 2, \quad \det B_I = 0 \quad (|I| = 3, 4). \quad (13)$$

There is a quadruple (x_0, x_1, x_2, x_3) for which all the finitely many relations in (13) hold simultaneously. This follows by taking a point in the intersection of the corresponding conull subsets of $\mathbb{R}^4$; the atomlessness of ν_R removes the diagonals. Write $b_{jk} = B(x_j, x_k)$.

The one- and two-minor equations give

$$b_{jj} = 2, \quad b_{jk}b_{kj} = 2 \quad (j \neq k). \quad (14)$$

In particular every off-diagonal entry is nonzero. Principal minors are unchanged by diagonal similarity. Conjugating the four-by-four matrix by a diagonal matrix, we may therefore arrange

$$b_{0j} = 1, \quad b_{j0} = 2 \quad (j = 1, 2, 3). \quad (15)$$

Fix distinct $j, k \in \{1, 2, 3\}$ and put $a = b_{jk}$. Then $b_{kj} = 2/a$, and the zero determinant of the principal submatrix on $\{0, j, k\}$ reads

$$0 = \det \begin{pmatrix} 2 & 1 & 1 \\ 2 & 2 & a \\ 2 & 2/a & 2 \end{pmatrix} = \frac{2(a^2 - 2a + 2)}{a}. \quad (16)$$

Hence $a \in \{1 + i, 1 - i\}$. Direct each edge $j \rightarrow k$ when $b_{jk} = 1 + i$; by (14), the reverse entry is then $1 - i$. For the three vertices 1, 2, 3, a directed three-cycle would give

$$\det \begin{pmatrix} 2 & 1 + i & 1 - i \\ 1 - i & 2 & 1 + i \\ 1 + i & 1 - i & 2 \end{pmatrix} = -8, \quad (17)$$

contrary to the three-minor equation. The tournament is therefore transitive. Relabeling x_1, x_2, x_3 if necessary, the full matrix is

$$M = \begin{pmatrix} 2 & 1 & 1 & 1 \\ 2 & 2 & 1 + i & 1 + i \\ 2 & 1 - i & 2 & 1 + i \\ 2 & 1 - i & 1 - i & 2 \end{pmatrix}. \quad (18)$$

A direct determinant calculation gives

$$\det M = -4. \quad (19)$$

This contradicts the zero four-point correlation in (13). No determinantal kernel exists, completing the proof of Theorem 1. $\square$

5 Verification, scope, and novelty

The accompanying exact-arithmetic script `code/verify_counterexample.py` checks the determinant factorization in (16), the transitive and cyclic three-by-three determinants, and $\det M = -4$. It also constructs Schur functions from the Newton recurrence and verifies the vanishing in (10) for every partition through total degree eight. The command used was

```
conda run --no-capture-output -n sandbox python \
code/verify_counterexample.py
```

and all exact checks passed. These checks are not substitutes for the all-degree mean-value proof or the matrix argument above.

The counterexample uses the continuous alternative $E = \mathbb{R}$ allowed by the source. It does not address whether a converse becomes true for a countable discrete E , for uncentered positive specializations, or under additional uniform-integrability or rigidity hypotheses. Those strengthenings are separate questions.

A bounded novelty search on 11 August 2026 checked the run indexes, the full source of arXiv:2111.05606, the cited Borodin–Olshanski–Strahov source arXiv:math-ph/0505021 [2], and arXiv/web queries for the exact conjecture and close variants. No later claimed resolution or this Cauchy counterexample was located. The detailed search record is in `novelty_search.md`; novelty confidence is moderate, not exhaustive.

Human-review recommendation. Review the circle pushforward and mean-value step, then check the normalization under a change of reference measure and the four-point principal-minor obstruction. These are the only substantive mechanisms; no numerical or unproved external input enters the counterexample.

References

- [1] A. I. Bufetov and P. Lazag, *A determinantal point process governed by an integrable projection kernel is Giambelli compatible*, arXiv:2111.05606 (2021).
- [2] A. Borodin, G. Olshanski, and E. Strahov, *Giambelli compatible point processes*, Adv. Appl. Math. 37 (2006), 209–248; arXiv:math-ph/0505021.